

Ina Câmpan

ENSAE Engineering Student (3rd year) – Dual Degree with UC3M (Madrid)

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Professional Experience

EMBL-EBI (European Bioinformatics Institute) – Research Trainee

Pathogen Informatics and Modelling Group, supervised by John Lees

June 2026 – August 2026

Hinxton, England

- Recipient of an **excellence scholarship from the French Embassy in the United Kingdom** (acceptance rate < 10%)
- Designed a **benchmarking pipeline** (Python, HPC/SLURM) comparing 9 gene clustering methods (CD-HIT, MMseqs2, DIAMOND, Panaroo, PPanGGOLiN...) on 6 simulated datasets (Infinitely Many Genes evolutionary model, with ground truth) and 50 real *Streptococcus pneumoniae* genomes
- Implemented **sketching** (k-mers, HDBSCAN) and **embedding**-based approaches (ProtT5), and introduced statistical agreement metrics between methods (ARI, AMI, permutation tests) to assess their significance
- Results presented as a poster at Romanian AI Days, September 2026

Collège de France – Research Assistant

Ecology and Evolution of Health team, supervised by Samuel Alizon and Olivier Supplisson

June 2025 – August 2025

Paris, France

- Developed and deployed an **R Shiny** application visualising the spatial epidemiology of genital ulcers caused by herpes simplex viruses (HSV) in France
- Integrated estimates from **Bayesian hierarchical spatial models** (INLA/inlabru) and **generalised additive models**

Education

Engineering Degree (Grande École Programme)

ENSAE (National School of Statistics and Economic Administration) - IP Paris

September 2024 – June 2027

Palaiseau, France

- Main courses** (GPA 4.0 - top 15%): Stochastic Processes, Probability Theory, Statistics, Optimisation, Econometrics, Surveys, Machine Learning, Monte Carlo Methods, Linear Time Series, Macroeconomics and Microeconomics
- Volunteering**: Vice-President of Fem'ENSAE (feminist association), Ticketing Coordinator (Arts Student Association)

MSc in Machine Learning for Health

UC3M (Universidad Carlos III de Madrid)

September 2026 – June 2027

Madrid, Spain

- Main courses**: Biosignals and Bioimaging, Statistical Signal Processing, Deep Learning, Computer Vision, Clinical NLP

Double Bachelor's Degree in Mathematics and Computer Science

Sorbonne University, Faculty of Science and Engineering

September 2021 – May 2024

Paris, France

- Ranked 1st** out of 500 students in first year (Formal Sciences track), graduated with **Highest Honours** (Mention Très Bien) over all three years, in both disciplines
- Main courses**: Measure Theory, Probability, Functional and Numerical Analysis, Linear and Bilinear Algebra, Data Science, Databases and PL/SQL, Data Structures, Introduction to Mathematical Epidemiology, Cryptology

Applied Statistics Project

Women at the Top: Who Reaches the Highest Salaries?

ENSAE - CREST, supervised by Bertrand Garbinti

October 2025 – May 2026

Palaiseau, France

- Analysis of the **representation of women** at the top of the wage distribution in France using **Base Tous Salariés** data (Insee 2022, 2.4 million observations)
- Oaxaca-Blinder decomposition**: the unexplained share of the gender pay gap rises from 23% (Bottom 90%) to over 80% (Top 10% and Top 1%)
- Logistic regression**: with comparable profiles, female executives are 8.5 percentage points less likely to reach the Top 10%

Skills

Programming Python (Pandas, NumPy, Scikit-learn, PyTorch), C, R (inlabru, INLA, Shiny), SAS, SQL, SLURM/HPC, UNIX, Gurobi, Git, LaTeX
Languages Bilingual in French and Romanian, Fluent in English, Spanish through immersion in Madrid

Interests

Design Website design, film photography

Sports Olympic Family Assistant at the Paris 2024 Olympic Games (GFS team), Fencing and fitness (as part of my studies at ENSAE)